TEST REPORT

Report No. : **ZKT-2601161132R**

Issue Date : Jan 22, 2026

Applicant name : **Guangzhou Xingyi Electronic Technology Co., Ltd.**

Address : 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.

Sample Description

Sample name : ATK-CLRK3562JB CORE BOARD

Trade mark : ALIENTEK

Model No. : ATK-CLRK3562JB

Manufacturer : Guangzhou Xingyi Electronic Technology Co., Ltd.

Address : 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.

Test Requested

No.	Requested	Conclusion
1	RoHS Directive 2011/65/EU Annex II amending Annex (EU)2015/863 and amending Annex (EU)2017/2102 —Lead, Cadmium, Mercury, Hexavalent Chromium, PBBs and PBDEs Content —Di-(2-ethylhexyl) phthalate(DEHP), Benzylbutyl phthalate(BBP), Dibutyl phthalate (DBP), Diisobutyl phthalate(DIBP) Content	PASS

Reviewer by : _____ Approved by : _____

Shenzhen ZKT Technology Co., Ltd. Approved

Testing Laboratory	:	**Shenzhen ZKT Technology Co., Ltd.**
Address	:	1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, ShenZhen, China.
Test Report Form No.	:	TF：2025-07
Received Date	:	Jan 16, 2026
Testing Period	:	Jan 16, 2026 to Jan 22, 2026

Version

Report No.	Issue Date	Description	Approved
ZKT-2601161132R	Jan 22, 2026	Original	Valid

Statement

1. The information as listed on the first page of this test report was all provided by the client except the received date, testing period, test result(s) and test request. The client shall be responsible for the representativeness of sample and authenticity of materials, for which Shenzhen ZKT Testing Technology Co., Ltd. shall bear no responsibilities.
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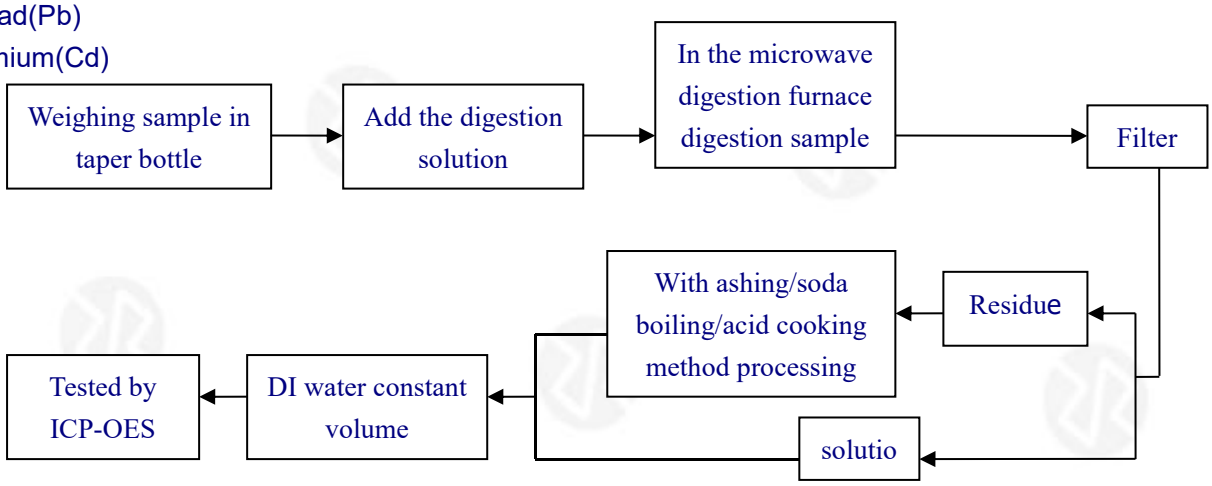
4. If there is any objection to the report, client shall lodge a complaint with the company on the 30 working days from the date of receipt of the report. No late application will be accepted.

Test Method:

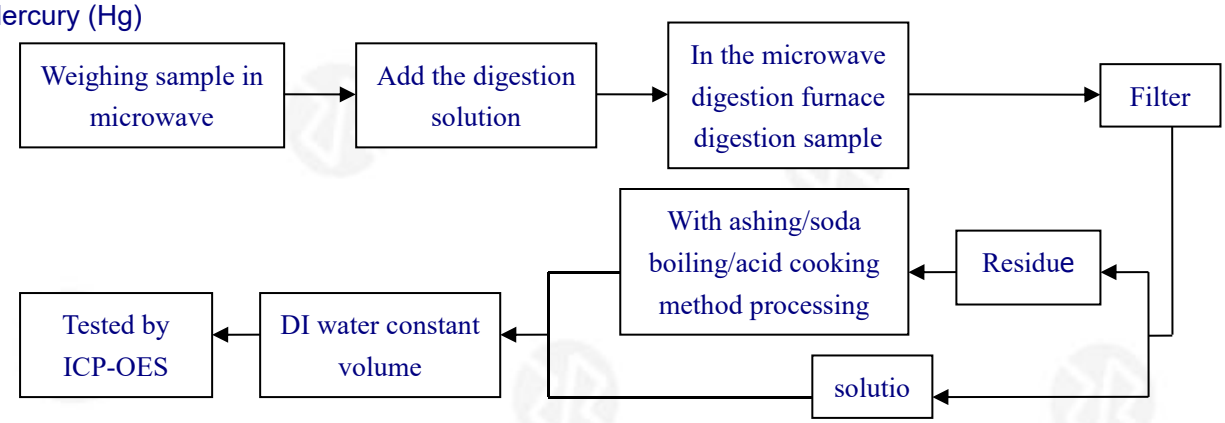
Test item(s)	Test Method(s)	Equipment(s)	MDL
Chemical Testing			
Mercury(Hg)	IEC 62321-4:2013+A1:2017	ICP-OES	2mg/kg
Lead(Pb)	IEC 62321-5:2013	ICP-OES	2mg/kg
Cadmium(Cd)	IEC 62321-5:2013	ICP-OES	2mg/kg
PBB	IEC 62321-6:2015	GC-MS	2mg/kg
PBDE			2mg/kg
Chromium(Cr VI) for colourless and coloured corrosion-protected coatings on metals	IEC 62321-7-1:2015	UV-Vis	0.10μg/cm2
Chromium(Cr VI) for polymers and electronics	IEC 62321-7-2:2017	UV-Vis	2mg/kg
DEHP	IEC 62321-8:2017	GC-MS	30mg/kg
DBP			30mg/kg
BBP			30mg/kg
DIBP			30mg/kg

Test Flow:

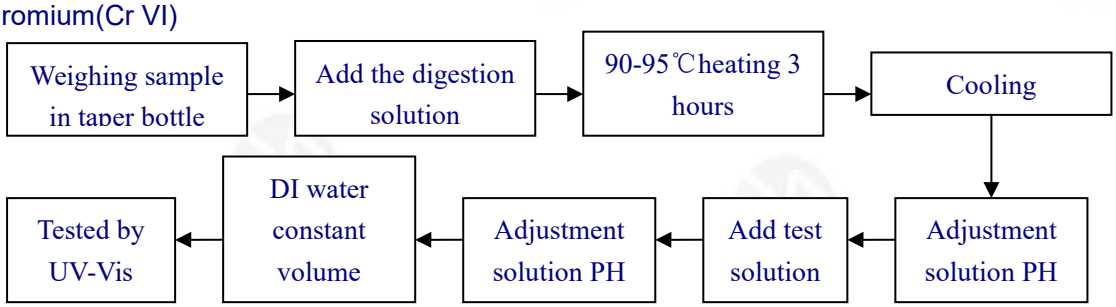
1. Lead(Pb)
Cadmium(Cd)



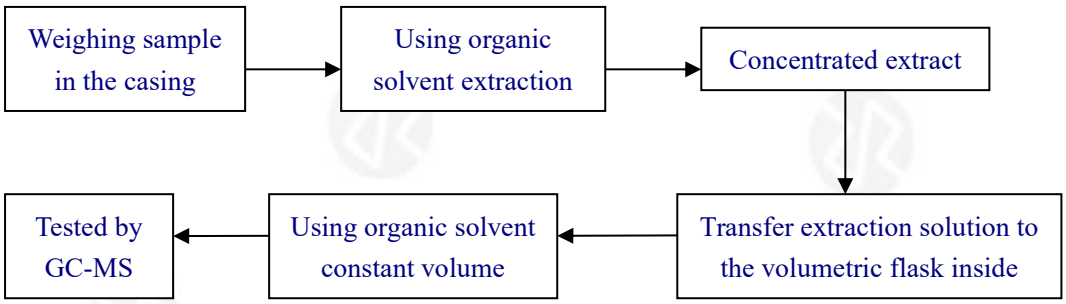
2. Mercury (Hg)



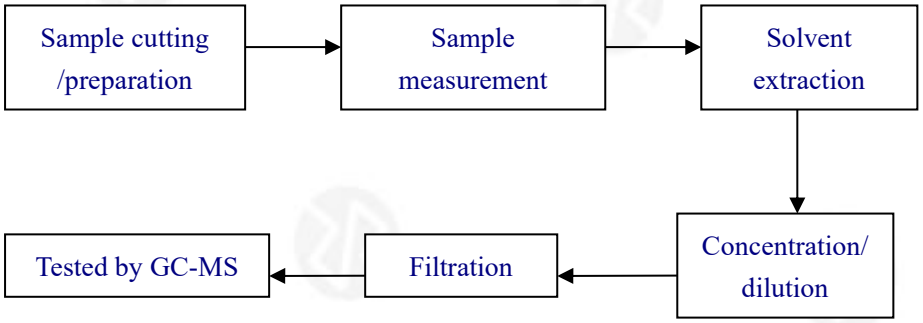
3. Chromium(Cr VI)



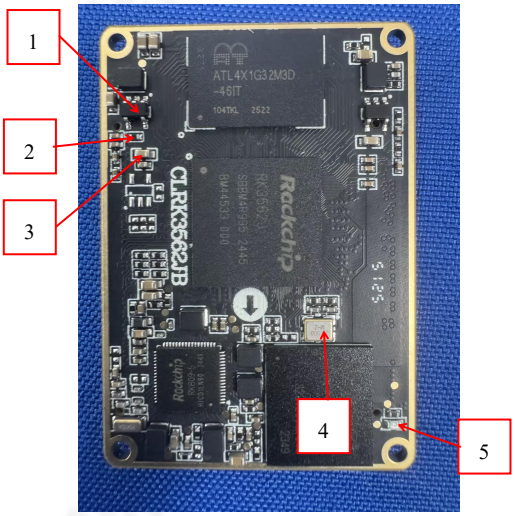
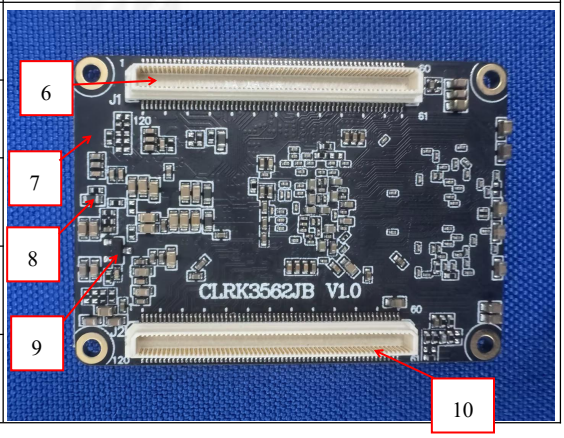
4. PBBs/ PBDEs



5. DEHP/ BBP/ DBP/ DIBP



Test Sample Description:

Sample No.	Description	Sample Photo
1	IC	
2	SMD Diode	
3	SMD Capacitor	
4	Crystal Oscillator	
5	SMD LED	
6	Beige plastic	
7	PCB	
8	SMD Resistor	
9	SMD Transistor	
10	Copper Metal	

Test Results:

RoHS screening test

Heavy metal (Cadmium Chromium Mercury Lead)content & Bromine content

Test Method : IEC 62321-3-1:2013

Analysis By : Energy Dispersive X-ray Fluorescence Spectrometers.

Sample No.	Total Cadmium	Total Lead	Total Mercury	Total Chromium	Total Bromine
1	BL	BL	BL	BL	BL
2	BL	BL	BL	BL	BL
3	BL	BL	BL	BL	BL
4	BL	BL	BL	BL	BL
5	BL	BL	BL	BL	BL
6	BL	BL	BL	BL	BL
7	BL	BL	BL	BL	BL
8	BL	BL	BL	BL	BL
9	BL	BL	BL	BL	BL
10	BL	BL	BL	BL	NA

Note:

1. All Concentrations express in “mg/kg” (milligram per kilogram)
2. mg/kg ~ ppm
3. “OL” denotes “over limit”
4. “BL” denotes “below limit”
5. “NA” denotes “Not Applicable”
6. “Inconclusive” denotes result is intermediate between “OL” and “BL”
7. “^”denotes the screening result was inconclusive(X) or over limit (OL)
8. thus further confirmation test was conducted results are listed in Heavy Metals Testand Flame retardants Test.

XRF screening limits for different materials:

Materials	Concentration (mg/kg)				
	Cd	Cr	Pb	Hg	Br
Metal	BL≤(70-3σ)<X<(130+3σ)≤OL	BL≤(700-3σ)<X	BL≤(700-3σ)<X<(1300+3σ)≤OL	BL≤(700-3σ)<X<(1300+3σ)≤OL	NA
Polymers	BL≤(70-3σ)<X<(130+3σ)≤OL	BL≤(700-3σ)<X	BL≤(700-3σ)<X<(1300+3σ)≤OL	BL≤(700-3σ)<X<(1300+3σ)≤OL	BL≤(300-3σ)<X
Composite material	BL≤(50-3σ)<X<(150+3σ)≤OL	BL≤(500-3σ)<X	BL≤(500-3σ)<X<(1500+3σ)≤OL	BL≤(500-3σ)<X<(1500+3σ)≤OL	BL≤(250-3σ)<X

Heavy Metals Test

Lead, Cadmium, Hexavalent Chromium and Mercury Tests

Test Method : IEC 62321-4:2013+A1:2017 &IEC 62321-5:2013 & IEC 62321-7-1:2015& IEC 62321-7-2:2017

Analysis By : ICP-OES & UV-VIS.

Element	**Total Cadmium** [mg/kg]	**Total Lead** [mg/kg]	**Total Mercury** [mg/kg]	**Hexavalent Chromium** [μg/cm2]	**Hexavalent Chromium** [mg/kg]
Detection Limit	5	5	5	0.10	5
Limit	100	1000	1000	0.10	1000

Note:

1. All Concentrations express in “mg/kg”(milligram per kilogram)
mg/kg ~ ppm.
2. “ND” = “Not Detected”.
3. Boiling-water-extraction:
   Negative = Absence of Cr(VI) coating / surface layer: the detected concentration in boiling-water-extraction solution is less than 0.10μg with 1cm2 sample surface area. Positive = Presence of Cr(VI) coating / surface layer: the detected concentration in boiling-water-extraction solution is greater than 0.13μg with 1cm2 sample surface area.
   Inconclusive =the detected concentration in boiling-water-extraction solution is greater than 0.10μg and less than 0.13μg with 1cm2 sample surface area.
4. Positive = result be regarded as not comply with RoHS requirement
   Negative = result be regarded as comply with RoHS requirement
5. “-” =Not regulated

Flame retardants Test

PBBs&PBDEs Content

Test Method : IEC 62321-6:2015

Analysis by : GC-MS

Test Item		Result [mg/kg]			RoHS Requirement [mg/kg]
		7	/	/	
PBBs	Monobromobiphenyl	< 5	< 5	< 5	-
	Dibromobiphenyl	< 5	< 5	< 5	-
	Tribromobiphenyl	< 5	< 5	< 5	-
	Tetrabromobiphenyl	< 5	< 5	< 5	-
	Pentabromobiphenyl	< 5	< 5	< 5	-
	Hexabromobiphenyl	< 5	< 5	< 5	-
	Heptabromobiphenyl	< 5	< 5	< 5	-
	Octabromobiphenyl	< 5	< 5	< 5	-
	Nonabromobiphenyl	< 5	< 5	< 5	-
	Decabromobiphenyl	< 5	< 5	< 5	-
	Sum of PBBs	< 5	< 5	< 5	< 1000
PBDEs	Monobromodiphenyl Ether	< 5	< 5	< 5	-
	Dibromodiphenyl Ether	< 5	< 5	< 5	-
	Tribromodiphenyl Ether	< 5	< 5	< 5	-
	Tetrabromodiphenyl Ether	< 5	< 5	< 5	-
	Pentabromodiphenyl Ether	< 5	< 5	< 5	-
	Hexabromodiphenyl Ether	< 5	< 5	< 5	-
	Heptabromodiphenyl Ether	< 5	< 5	< 5	-
	Octabromodiphenyl Ether	< 5	< 5	< 5	-
	Nonabromodiphenyl Ether	< 5	< 5	< 5	-
	Decabromodiphenyl Ether	< 5	< 5	< 5	-
	Sum of PBDEs	< 5	< 5	< 5	< 1000

Note:

1. All Concentrations express in “mg/kg” (milligram per kilogram)
2. mg/kg ~ ppm.
3. “<” denotes less than

Phthalate Test

DEHP&BBP&DBP&DIBP Content

Test Method : IEC 62321-8:2017

Analysis By : GC-MS.

Element	Di-(2-ethylhexyl) phthalate (DEHP) [mg/kg]	Benzylbutyl phthalate (BBP) [mg/kg]	Dibutyl phthalate (DBP) [mg/kg]	Diisobutyl phthalate(DIBP) [mg/kg]
Detection Limit	**50**	**50**	**50**	**50**
Limit	**1000**	**1000**	**1000**	**1000**
1	ND	ND	ND	ND
2	ND	ND	ND	ND
3	ND	ND	ND	ND
4	ND	ND	ND	ND
5	ND	ND	ND	ND
6	ND	ND	ND	ND
7	ND	ND	ND	ND
8	ND	ND	ND	ND
9	ND	ND	ND	ND

Note:

1. All Concentrations express in “mg/kg”(milligram per kilogram)
2. mg/kg ~ ppm.
3. “ND” = “Not Detected”.

Submitted Sample Photo

Photo 1

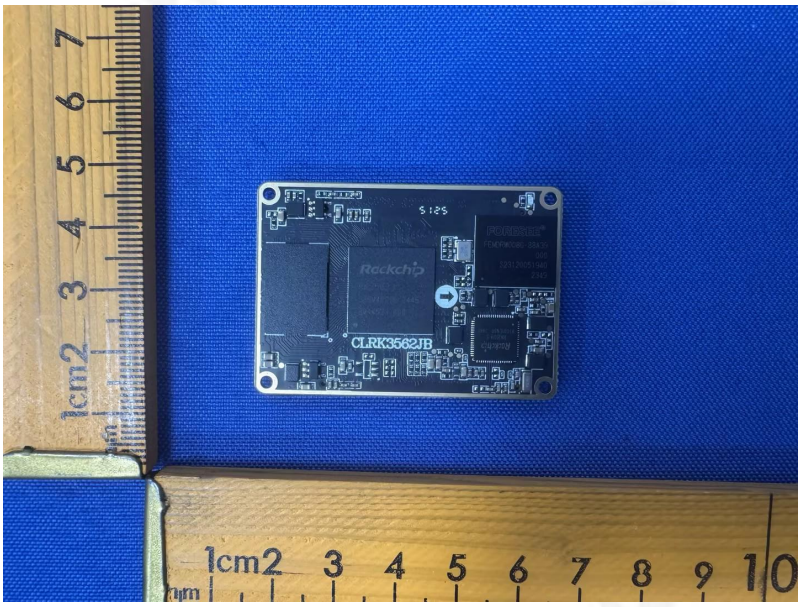
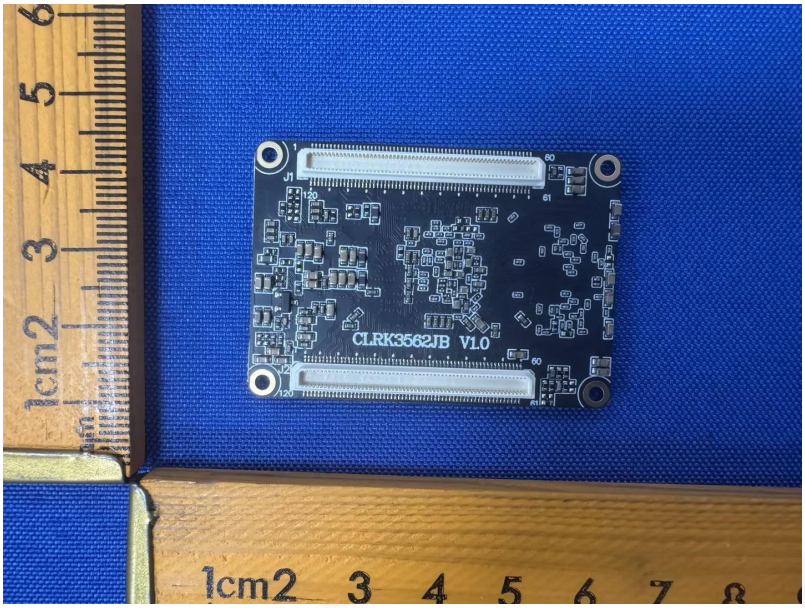


Photo 2



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